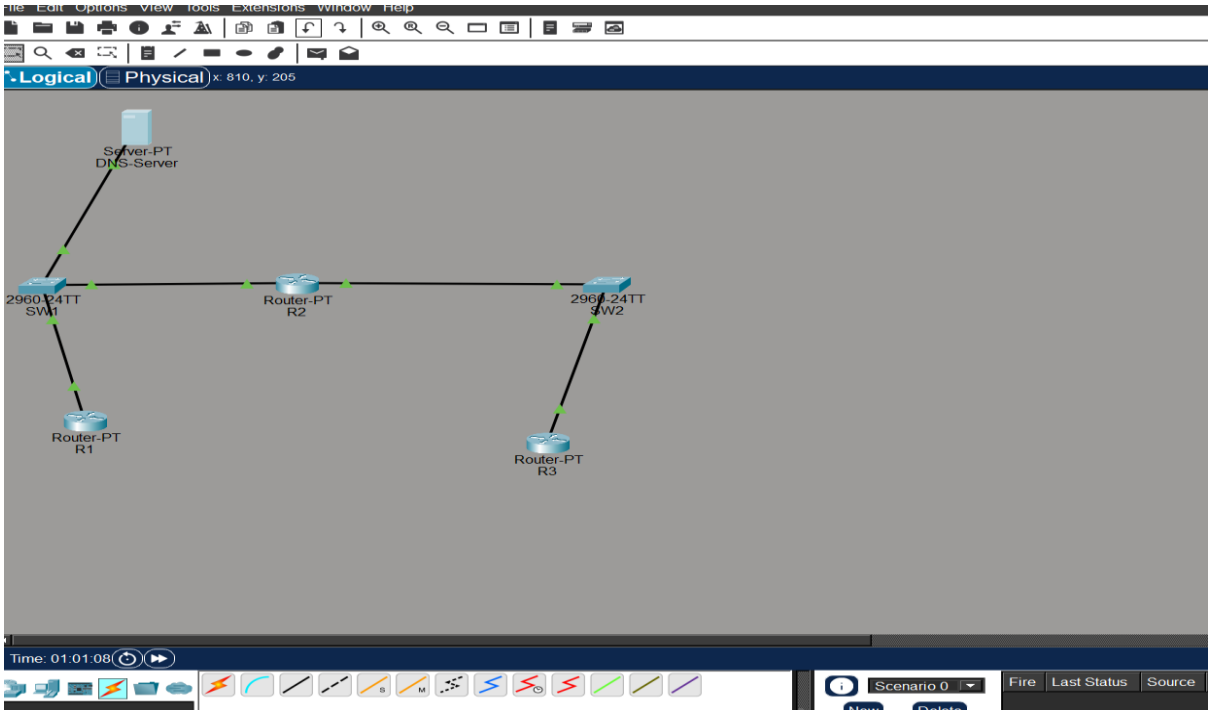
Week 3: Life of a Packet (10 Marks)



Configure the routers as DNS clients, pointing to the DNS server in the topology.

```
Router#show running-config
Building configuration...

Current configuration : 850 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$QOTvWLvJFe/Qd4Rxcf57d.
enable password enable password
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
!
!
!
ip name-server 10.10.10.10
!
!
!
```

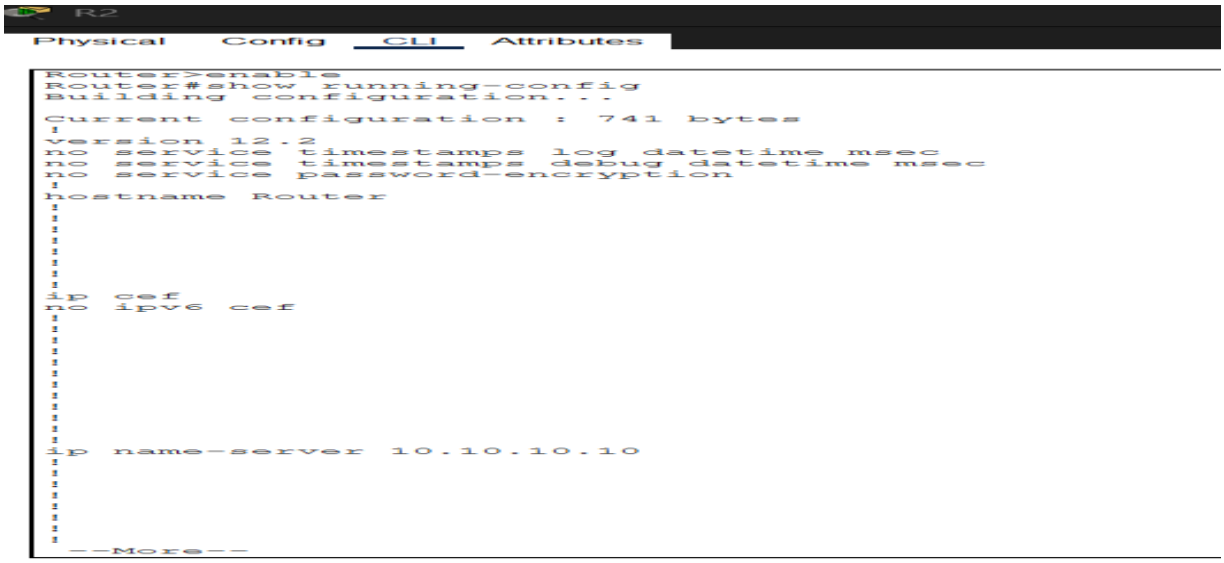
R1

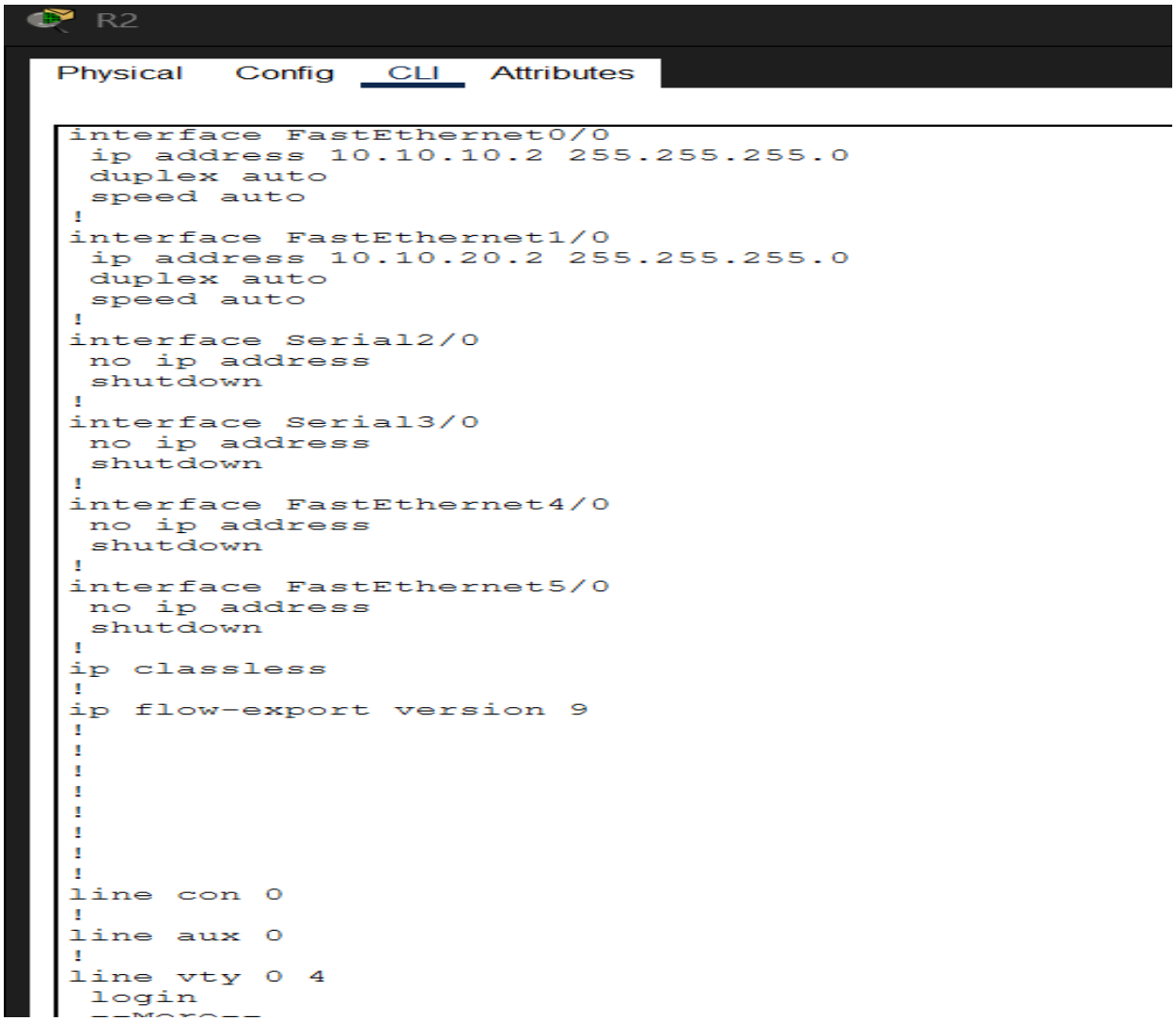
```
Router#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router#
```

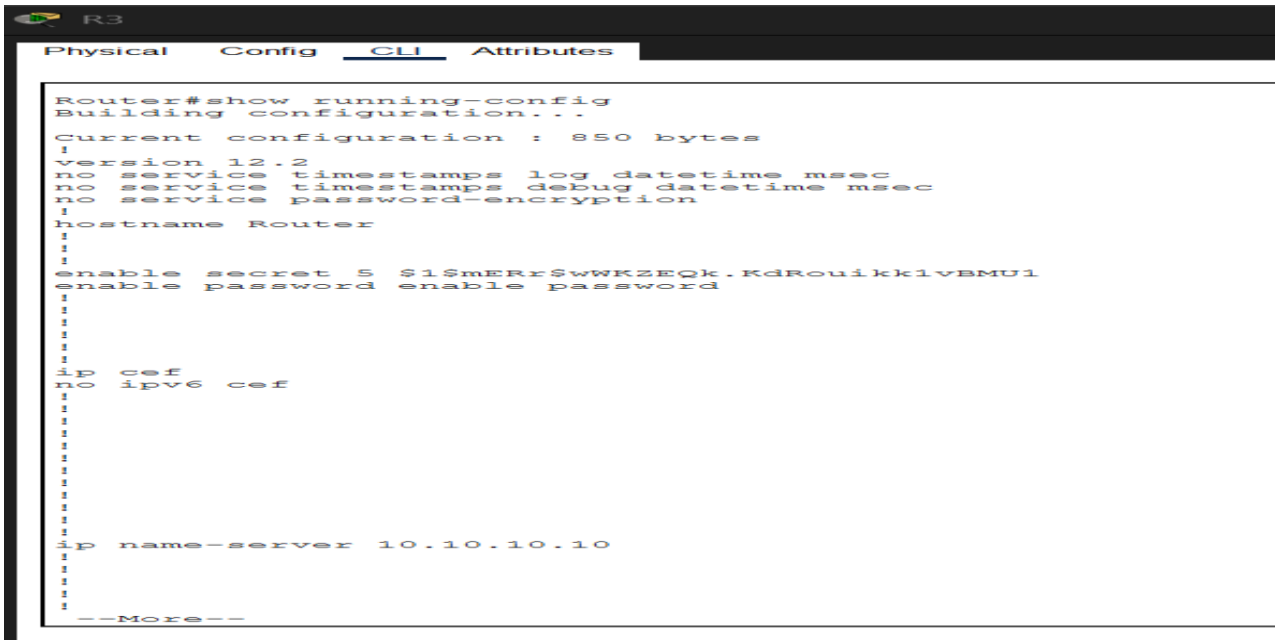
R2


R2
Physical
Config
CLI
Attributes
Router>enable
Router#show running-config
Building configuration...
Current configuration : 741 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
ip name-server 10.10.10.10
!
!
!
!
!
!
--More--


R2
Physical
Config
CLI
Attributes
interface FastEthernet0/0
ip address 10.10.10.2 255.255.255.0
duplex auto
speed auto
!
interface FastEthernet1/0
ip address 10.10.20.2 255.255.255.0
duplex auto
speed auto
!
interface Serial2/0
no ip address
shutdown
!
interface Serial3/0
no ip address
shutdown
!
interface FastEthernet4/0
no ip address
shutdown
!
interface FastEthernet5/0
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login


R2
Physical
Config
CLI
Attributes
!
interface Serial3/0
 no ip address
 shutdown
!
interface FastEthernet4/0
 no ip address
 shutdown
!
interface FastEthernet5/0
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end
Router#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
Router#

R3


R3
Physical
Config
CLI
Attributes
Router#show running-config
Building configuration...
Current configuration : 850 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
enable secret 5 1mERr$wWKzEQk.KdRouikk1vBMU1
enable password enable password
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
!
!
!
!
!
ip name-server 10.10.10.10
!
!
!
!
 --More--





1. Test name resolution from a router by pinging a hostname instead of an IP address.

R1


R1
Physical
Config
CLI
Attributes
shutdown
!
interface FastEthernet5/0
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
Router#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
Router#ping server.com
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
Router#

R2

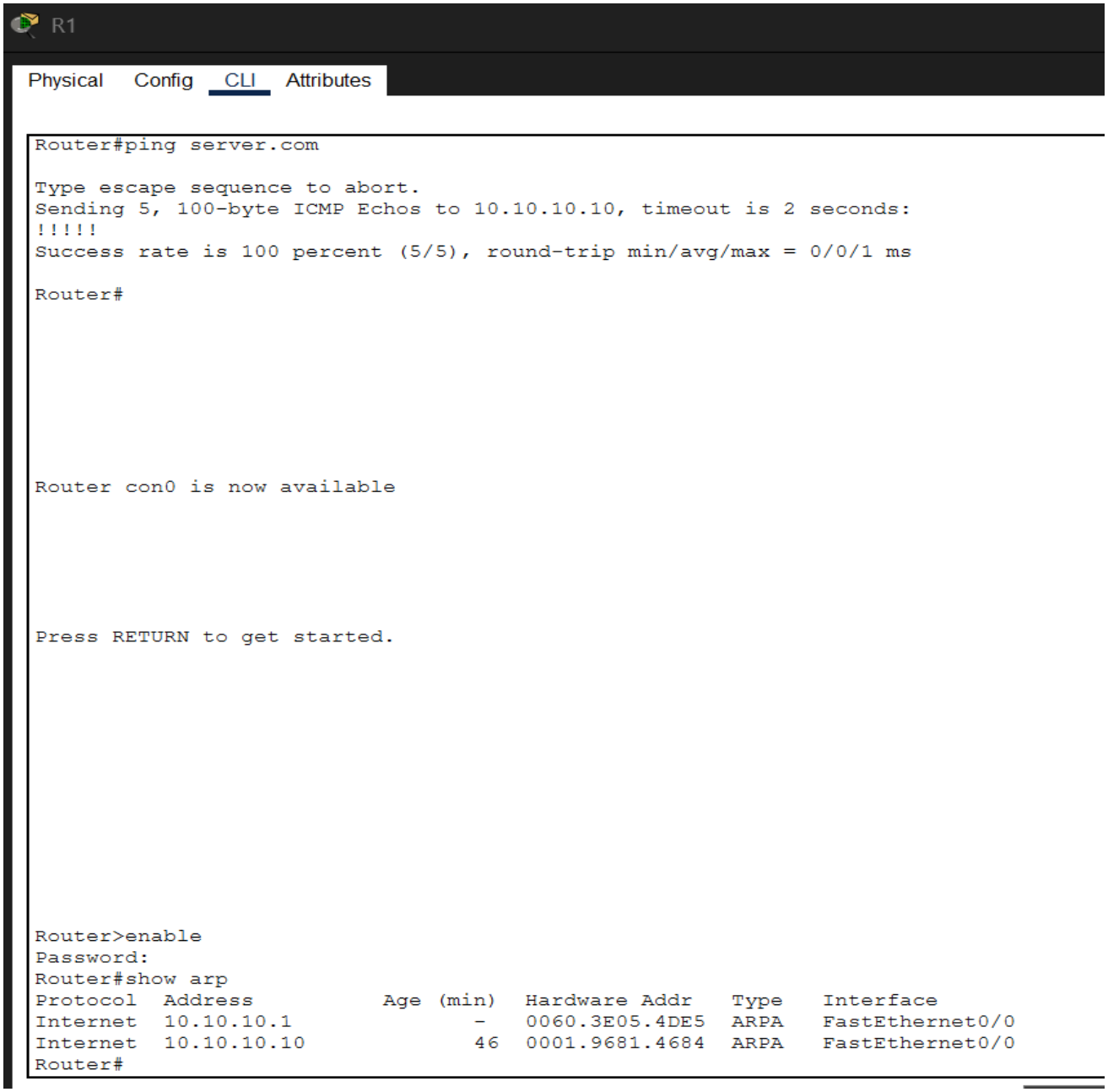

R2
Physical
Config
CLI
Attributes
shutdown
!
interface FastEthernet5/0
no ip address
shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
!
!
end
Router#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
Router#ping server.com
Translating "server.com"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
Router#

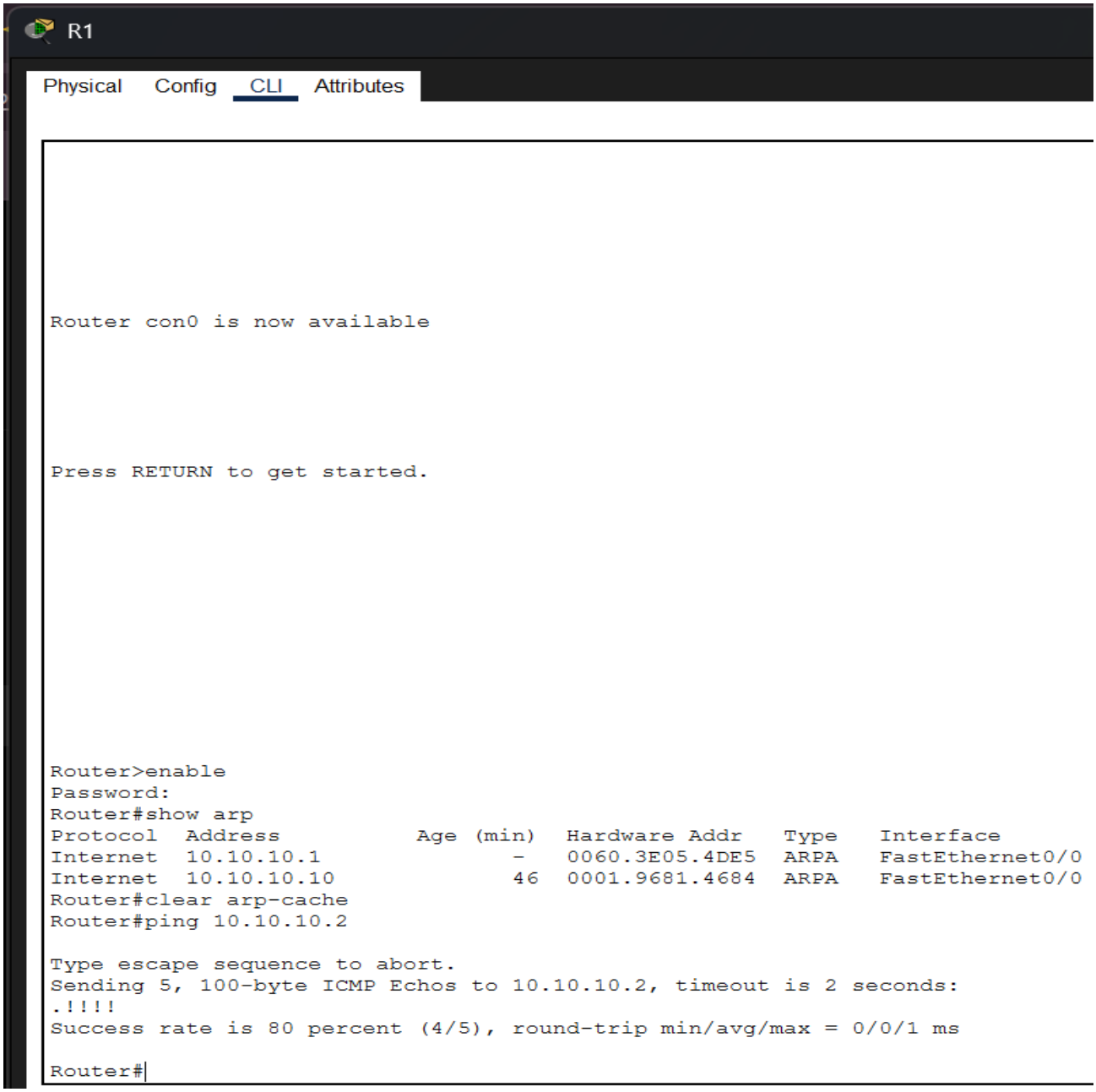
R3



1. Clear the ARP cache on a router, then generate traffic (e.g. a ping) to a directly connected neighbour.

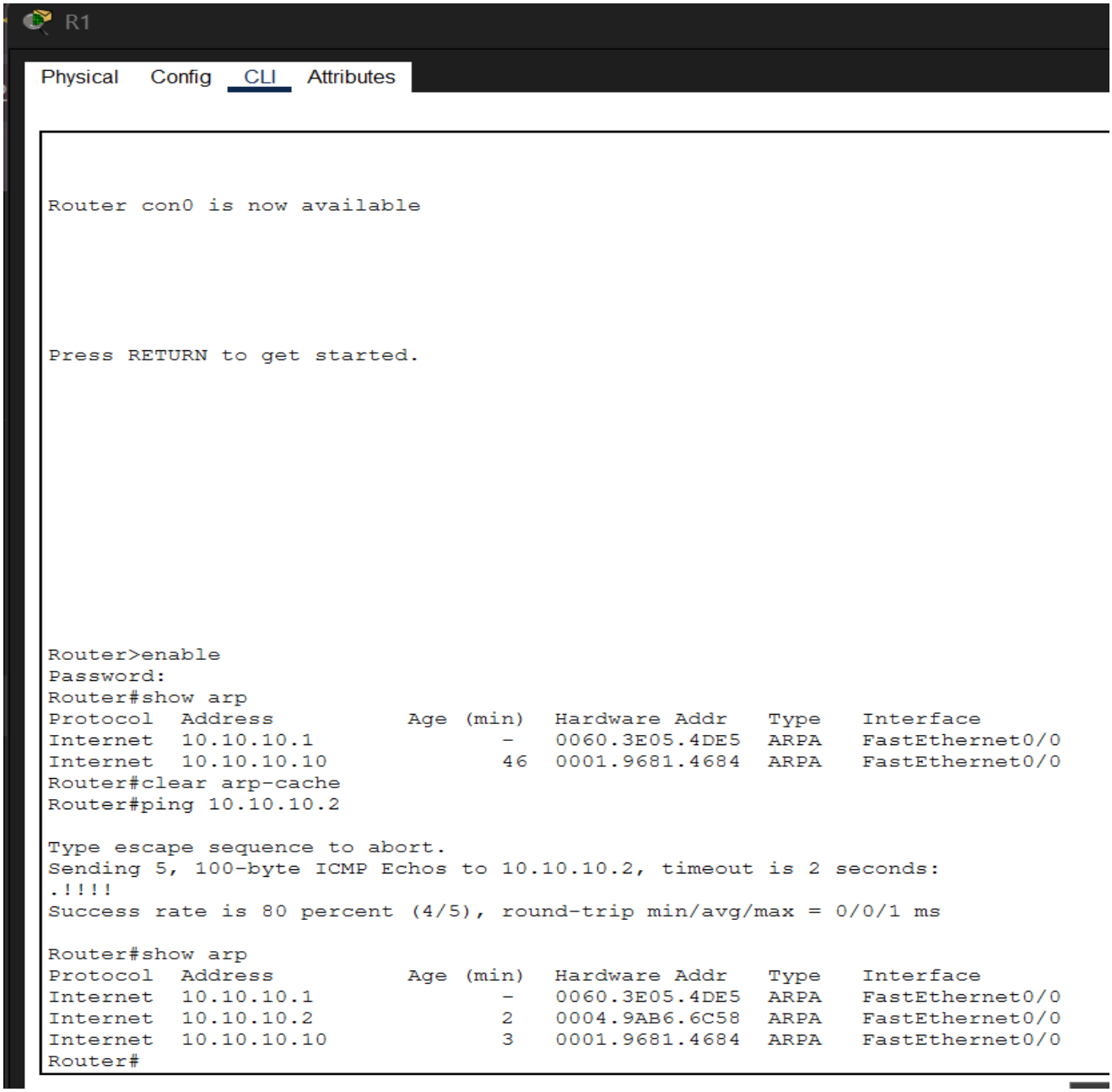
R1


R1
Physical
Config
CLI
Attributes
Router#ping server.com
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
Router#
Router con0 is now available
Press RETURN to get started.
Router>enable
Password:
Router#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 10.10.10.1 - 0060.3E05.4DE5 ARPA FastEthernet0/0
Internet 10.10.10.10 46 0001.9681.4684 ARPA FastEthernet0/0
Router#



1. Examine the ARP cache on the routers with 'show arp' and identify the newly learned entry.

R1



Reflection Questions – Week 3: Life of a Packet

1. What information does an ARP entry give you that a routing table entry does not?

An Address Resolution Protocol (ARP) entry provides the MAC address associated with an IPv4 address on a directly connected network. It allows a router to identify the physical network address of a device so that data can be delivered to the correct device over an Ethernet network. An ARP table usually contains the IP address, the corresponding MAC address, and the interface or connection through which the device can be reached.

A routing table, on the other hand, contains information about the paths that packets can take to reach different destination networks. It includes destination network addresses,

subnet masks, next-hop IP addresses, and outgoing interfaces. However, the routing table does not normally contain the MAC address needed to deliver an Ethernet frame to the next device.

For example, if a router needs to send a packet to a computer with IP address 192.168.1.10, its routing table helps it determine which interface to use. The ARP table helps it find the MAC address associated with that IP address on the local network. Therefore, the main difference is that the routing table determines where to forward a packet, while the ARP table helps determine the destination MAC address for delivery on the local network.

2. Why does a router need to perform an ARP lookup even after it already knows the correct route from its routing table?

A router needs to perform an ARP lookup because knowing the correct route does not automatically provide the MAC address required to transmit an Ethernet frame. The routing table helps the router determine the best path toward the destination network, but the router still needs the next-hop device's MAC address to forward the packet over an Ethernet connection.

When a router receives a packet, it examines the destination IP address and checks its routing table to select the appropriate route. If the packet must be forwarded through an Ethernet interface, the router checks its ARP table to determine whether it already knows the MAC address of the next-hop device. If the address is not available, the router sends an ARP request to discover it. The device with the matching IP address responds with its MAC address, and the router stores this information in its ARP cache for future use.

Once the MAC address is known, the router encapsulates the IP packet inside a new Ethernet frame and forwards it to the next hop. The router may repeat this process at each router along the path when forwarding over Ethernet networks. The destination IP address generally remains the same throughout routing, while the Ethernet frame's source and destination MAC addresses change at each routed hop.

For example, if Router R1 needs to forward a packet to Router R2, its routing table may indicate that R2 is the next hop. However, R1 still needs R2's MAC address to construct the Ethernet frame. ARP provides that missing information, allowing the packet to be transmitted successfully.

In conclusion, routing and ARP perform different but complementary functions. Routing determines the path a packet should follow, while ARP resolves the next-hop IPv4 address to a MAC address so the router can deliver the packet over the local network. Without the required MAC address, the router cannot transmit the Ethernet frame, even if it has already identified the correct route.